Question	Answer	Marks	Partial Marks
14(b)	$\frac{x+4}{3}$ oe	**2**	**M1** for correct first step $y + 4 = 3x$ or $x = 3y - 4$ or $\frac{y}{3} = x - \frac{4}{3}$
14(c)	$a = 12$, $b = -1$	**2**	**B1** for either a or b correct or **M1** for $3(4x + 1) - 4$
14(d)	$\frac{22-7x}{(3x-4)(4x+1)}$ or $\frac{22-7x}{12x^2-13x-4}$ final answer	**3**	**B1** for $2(4x + 1) - 5(3x - 4)$ oe or better isw **B1** for common denominator $(3x - 4)(4x + 1)$ oe isw
15(a)	$17 - 7\sqrt{5}$ final answer	**2**	**B1** for 3 correct terms from $2 - 6\sqrt{5} - \sqrt{5} + 3 \times \sqrt{5} \times \sqrt{5}$ oe
15(b)	$\frac{3\sqrt{10}}{5}$ cao	**2**	**M1** for $\frac{6}{\sqrt{10}} \times \frac{\sqrt{10}}{\sqrt{10}}$ oe
16	$3x^3 + 5x^2 - 34x - 24$ final answer	**3**	**B2** for correct expansion unsimplified or simplified four-term expression of correct form with 3 terms correct or **B1** for one pair of brackets expanded with at least 3 terms out of 4 correct
17(a)	$\frac{9}{100}$ oe	**2**	**M1** for $\frac{3}{10} \times \frac{3}{10}$
17(b)(i)	$\frac{2}{5}$ $\frac{4}{5}$ $\frac{1}{5}$	**2**	**B1** for $\frac{2}{5}$
17(b)(ii)	$\frac{16}{30}$ oe	**3**	**3FT** *their* tree diagram dep on probabilities < 1 **M2FT** for $\frac{4}{6} \times \frac{2}{5} + \frac{2}{6} \times \frac{4}{5}$ or **M1FT** for $\frac{4}{6} \times \frac{2}{5}$ or for $\frac{2}{6} \times \frac{4}{5}$
18(a)	$\frac{12}{x}$	**1**	
18(b)	$\frac{10}{x-4}$	**1**	

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Question	Answer	Marks	Partial Marks
18(c)	*their* $\frac{10}{x-4}$ – *their* $\frac{12}{x}$ = 1 oe	**M1**	
	$10x - 12x + 48 = x^2 - 4x$	**M2**	Correctly multiplying *their* brackets **and** clearing algebraic fractions e.g. $(x-4)\left(\frac{12}{x}+1\right)=10$ leading to $12-\frac{48}{x}+x-4=10$ and then $12x-48+x^2-4x=10x$ or **M1** for correctly clearing, or correctly collecting into a single fraction, two fractions both with different algebraic denominators e.g. $10x - 12(x - 4) = x(x - 4)$ or $\frac{10x-12(x-4)}{x(x-4)}$ [= 1]
	Leading to $0 = x^2 - 2x - 48$	**A1**	With no errors or omissions seen, dep on M3
18(d)	$(x + 6)(x - 8)$	**M2**	**M1** for $x(x - 8) + 6(x - 8)$ or $x(x + 6) - 8(x + 6)$ or $(x + a)(x + b)$ where $ab = -48$ or $a + b = -2$
	–6, 8	**B1**	
18(e)	1.5 or $1\frac{1}{2}$	**1**	FT $\frac{12}{\textit{their}\ 8}$
19	$\frac{1}{9}$	**2**	**M1** for 9^{-1} or for $\frac{1}{3^2}$ or $\frac{1}{\sqrt[3]{729}}$
20	$6\sqrt{3}$ oe	**4**	**B1** for $\tan 30 = \frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$ or for $\sin 60 = \frac{\sqrt{3}}{2}$ **and** $\sin 30 = \frac{1}{2}$ **M2** for $\frac{6}{\tan 30}$ or $\frac{6\sin(60)}{\sin(30)}$ or **M1** for $\frac{6}{x} = \tan 30$ or $\frac{\sin 60}{x} = \frac{\sin 30}{6}$
21(a)(i)	2**m** – **n**	**1**	
21(a)(ii)	2**m** + **n**	**1**	